

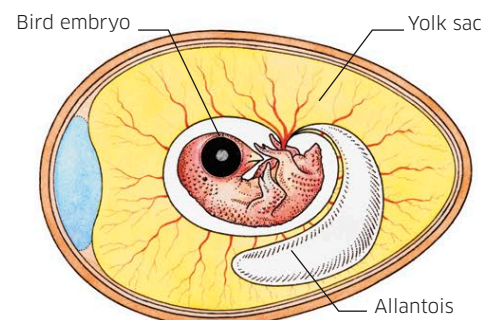
Producing young

The drive to reproduce is one of the most basic instincts in all animals. Many species devote their entire lives to finding a mate and making new young.

The most common way for animals to reproduce is through sexual reproduction—where sperm cells produced by a male fertilize egg cells produced by a female. The fertilized egg then becomes an embryo that will slowly grow and develop into a new animal. Many underwater animals release their sperm and eggs together into open water, but land animals must mate so that sperm are passed into the female's body and can swim inside it to reach her eggs.

Laying eggs on land

In some land animals, such as birds and reptiles, eggs are fertilized inside the mother's body and then laid—usually into a nest. These eggs have a hard, protective shell that encases the embryo inside and stops it from drying out. They also contain a big store of food—the yolk—which nourishes the embryo as it develops. It can take weeks or even months before the baby is big enough to hatch and survive in the world outside.



Inside a bird's egg

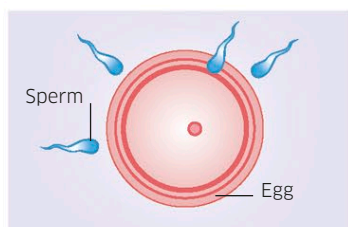
The shell of a bird's egg lets in air to help the embryo breathe. The yolk sac provides nutrients as it grows into a chick, while another sac, the allantois, helps collect oxygen and waste.

Giving birth to live young

Except for a few egg-laying species (called monotremes), mammals give birth to live young. The mother must support the growing embryos inside her body—a demanding task that may involve her taking in extra nutrients. The babies grow in a part of the mother's body called the uterus, or womb, where a special organ called a placenta passes them food and oxygen.

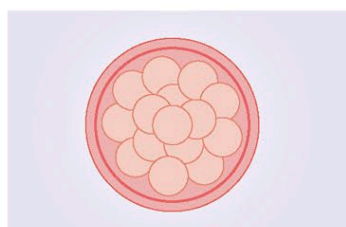
A new generation of mice

Some mammals, such as humans, usually give birth to one baby at a time. Others have large litters—like mice, which can produce up to 14 babies at one time. Each one starts as a fertilized egg, grows into an embryo, and then is born just three weeks later.



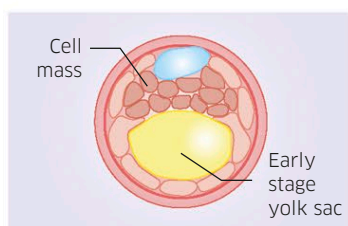
1 Fertilization

When a male mouse mates with a female, thousands of sperm enter her body and swim to her eggs. The first to arrive penetrates an egg—fertilizing it.



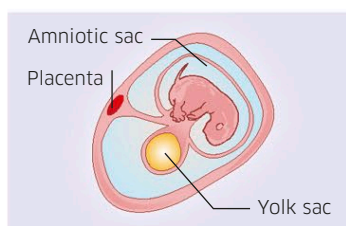
2 Embryo forms

The fertilized egg cell now contains a mixture of genes from the sperm and the egg. It divides multiple times to form a microscopic ball of cells called an embryo.



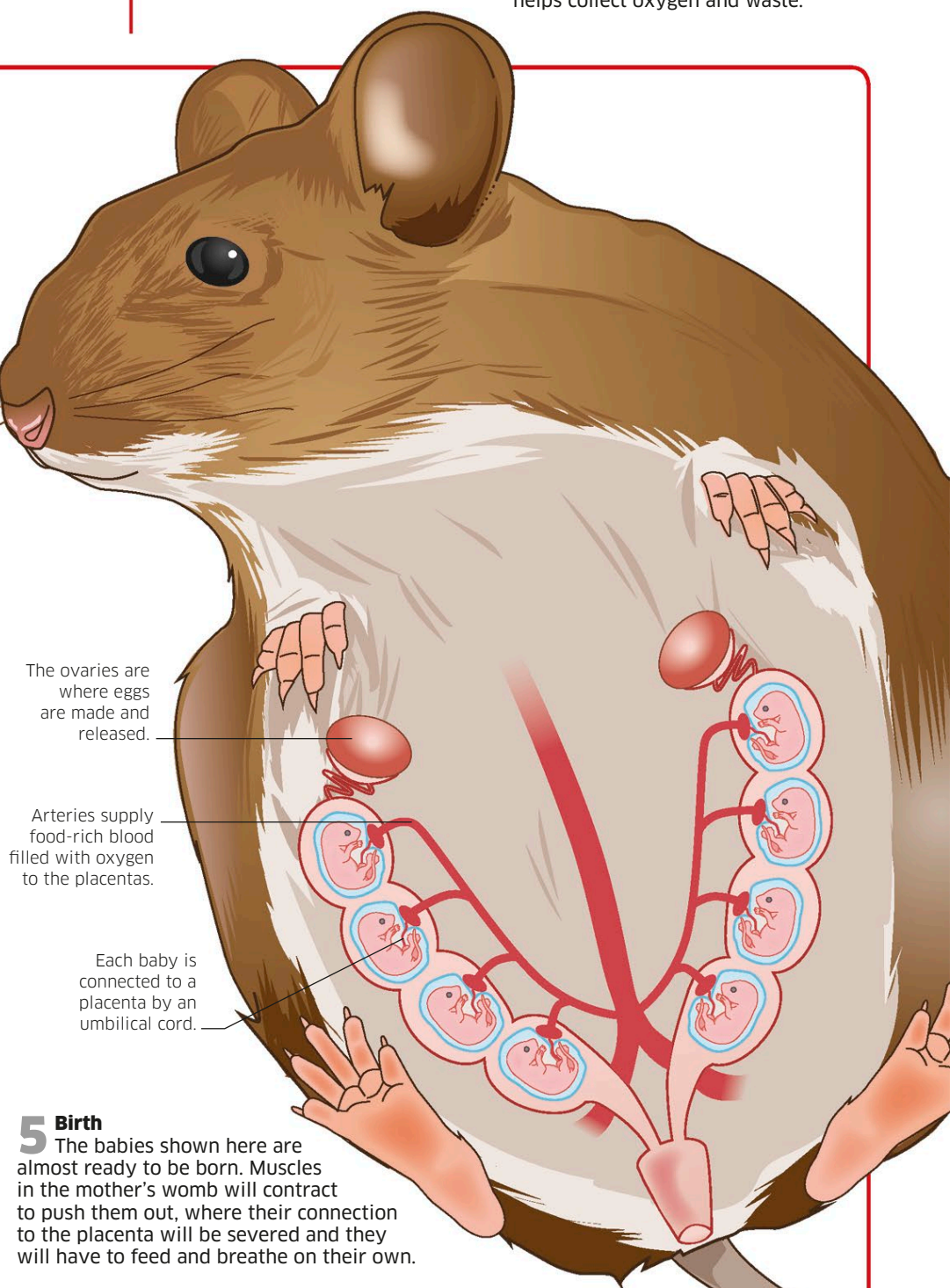
3 Implantation

The embryo becomes a hollow ball. A cell mass on one side will become the mouse's body. The ball travels into the womb to embed into its wall—an event called implantation.



4 Placenta grows

The baby mouse begins to form and gets nutrients from first a temporary yolk sac and then a placenta. A fluid-filled bag, the amniotic sac, cushions the embryo.



5 Birth

The babies shown here are almost ready to be born. Muscles in the mother's womb will contract to push them out, where their connection to the placenta will be severed and they will have to feed and breathe on their own.